No.	Status	Date
LNDE-RPT-20260803-034-DS-4-RA-101	Issued	2026-08-03

UT Thickness Data Sheet - 4-RA-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for the rich amine line 4-RA-101 located within the ATU-100. The scope is limited to the baseline thickness recording of replacement spools installed during the QSCP Turnaround 2026 per the applicable MOC.

Marcus Chen, acting as the lead UT technician, is responsible for the execution of the measurements and the accuracy of the recorded data. All activities were performed in accordance with the requirements of the Hamilton Plant asset integrity framework.

Technical Basis

Thickness measurements were performed under QSCP-NDT-001 Rev 4, utilizing the ATU-100 specific guidelines for ultrasonic thickness measurement. The inspection methodology adheres to the requirements of ASTM E797.

The measurements recorded herein serve as the primary data source for the baseline report LNDE-RPT-20260803-034-R0. All data points were captured to establish the initial wall thickness of the new A106 Gr.B spools.

Line Specifications

The subject line, 4-RA-101, has a nominal outer diameter of 114.3 mm and a nominal wall thickness of 8.56 mm. The design pressure is 862 kPa(g) and the design temperature is 82 °C.

The minimum allowable thickness (t-min) for this line is 3.81 mm. Any reading below this threshold is subject to the site NCR process.

Measurement Parameters

Data collection focused on the 304L CMLs. In accordance with the material specifications for stainless steel components, measurements were taken using the stainless velocity setting on LNDE-SS-003.

The resulting thickness values are integrated into the baseline dataset to facilitate future corrosion rate calculations and remaining life assessments.

Thickness readings, measured 2026-08-03

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2025-08-15	6.32	6.23	0.075	0.155	15.6	2029-08-03	
4-RA-101-02	8.56	3.81	2025-08-15	6.40	6.23	0.076	0.144	16.8	2029-08-03	
4-RA-101-03	8.56	3.81	2025-08-15	5.42	5.12	0.115	0.222	5.9	2029-07-17	gauge photo attached
4-RA-101-04	8.56	3.81	2025-08-15	6.82	6.65	0.065	0.144	19.7	2029-08-03	
4-RA-101-05	8.56	3.81	2025-08-15	6.20	6.05	0.084	0.175	12.8	2029-08-03	
4-RA-101-06	8.56	3.81	—	—	8.56	—	—	—	2027-02-01	
4-RA-101-07	8.56	3.81	2025-08-15	7.10	7.07	0.050	0.111	29.4	2029-08-03	
4-RA-101-08	8.56	3.81	2025-08-15	4.85	4.54	0.136	0.260	2.8	2027-12-27	gauge photo attached
4-RA-101-09	8.56	3.81	2025-08-15	7.04	6.97	0.053	0.103	30.6	2029-08-03	
4-RA-101-10	8.56	3.81	2025-08-15	6.86	6.57	0.069	0.160	17.3	2029-08-03	
4-RA-101-11	8.56	3.81	2025-08-15	5.34	5.18	0.115	0.229	6.0	2029-07-29	gauge photo attached
4-RA-101-12	8.56	3.81	2025-08-15	5.95	5.77	0.185	0.165	10.6	2029-08-03	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2025-11-03	3.52	3.59	0.011	0.010	>50	2032-08-03	
2-RF-102-02	3.91	2.16	2025-11-03	3.49	3.49	0.013	0.024	>50	2032-08-03	
2-RF-102-03	3.91	2.16	2025-11-03	3.64	3.66	0.009	0.003	>50	2032-08-03	
2-RF-102-04	3.91	2.16	—	—	3.87	—	—	—	2027-08-03	
2-RF-102-05	3.91	2.16	—	—	3.90	—	—	—	2027-08-03	
2-RF-102-06	3.91	2.16	—	—	3.96	—	—	—	2027-08-03	
2-RF-102-07	3.91	2.16	—	—	3.87	—	—	—	2027-08-03	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2026-08-03	_____